STA 35C: Statistical Data Science III

Lecture 6: Multiple Linear Regression & Polynomial Regression

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Recap: Simple linear regression

- Linear model: $Y = \beta_0 + \beta_1 X + \varepsilon$
- *Least squares estimation* chooses $(\hat{\beta}_0, \hat{\beta}_1)$ minimizing

$$\mathrm{RSS} = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 = \sum_{i=1}^{n} \left(\hat{\beta}_0 + \hat{\beta}_1 x_i - y_i\right)^2$$

- *Interpretation:* β_1 is the average change in the mean response per one-unit increase in X
  - Associational, not necessarily causation
- *Inference:* $\hat{\beta}_0$ and $\hat{\beta}_1$ are random variables
  - Standard errors, confidence intervals, and tests
  - Rejecting $H_0 : \beta_i = 0$ gives evidence of association
- Model fit is summarized by RSE and

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} \qquad \text{where} \qquad \mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

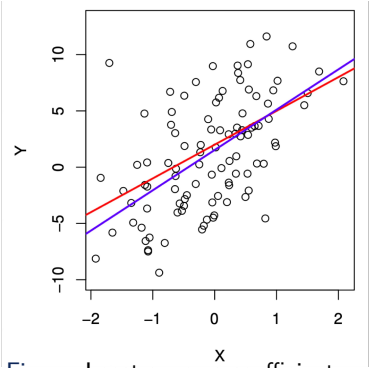

Figure: Least squares coefficient estimates from 100 data points. Red: population regression line, Blue: least squares line [JWHT21, Figure 3.3].

Agenda

Last time: Simple linear regression

- Model, least squares, inference, and model fit

Today: Extending simple linear regression

- Multiple linear regression
  - Motivation: *"What if we have more than one predictor?"* $X_1, X_2, \ldots$
  - Model
  - Estimation
  - Inference
  - Model fit
  - Prediction intervals
- Polynomial regression
  - Motivation: *"What if X_1 and X_2 interact, or if Y depends on X^2 instead of X?"*

Motivation for multiple linear regression

Recall the `Advertising` dataset and the three separate simple linear regression lines:

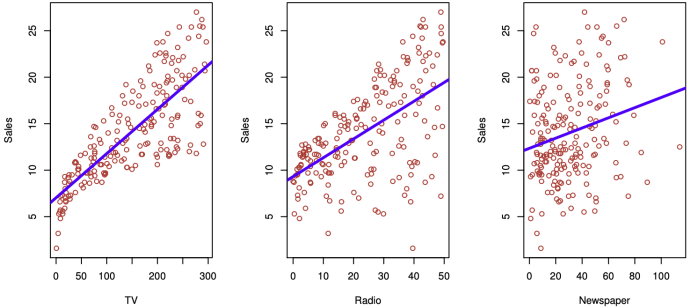

Figure: The `Advertising` data set: `Sales` of a product in 200 different markets against advertising budgets for three media: `TV`, `Radio`, and `Newspaper` [JWHT21, Figure 2.1].

Problem: Each simple linear regression line ignores the other two predictors

Question: Can we extend our analysis to accommodate *all* predictors simultaneously?

Multiple linear regression: Setup

We predict Y using multiple variables $X_1, X_2, \ldots, X_p$, assuming:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon,$$

- *Model parameters*: $\beta_0, \beta_1, \ldots, \beta_p$ are fixed, unknown constants
- ε is an error term, independent of $X_1, \ldots, X_p$

The coefficient β_j is interpreted as the the **average change** in the **mean response** for a one-unit increase in X_j, **holding all other predictors fixed**

- This is an associational interpretation, not necessarily a causal one

Once we *estimate* $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ from training data, we can predict

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Visualizing multiple linear regression

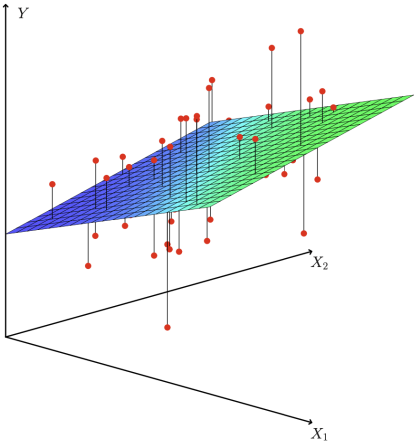

Figure: An illustration of multiple linear regression [JWHT21, Figure 3.4]. With two predictors, the fitted surface is a plane; with many predictors, it is a hyperplane.

Coefficient estimation via least squares

Coefficients $\beta_0, \beta_1, \ldots, \beta_p$ must be estimated from data $(x_1, y_1), \ldots, (x_n, y_n)$, where

$$x_i = (x_{i1}, x_{i2}, \ldots, x_{ip}).$$

Again, we use the **least squares** criterion:

- The *least squares* approach chooses $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ to minimize the RSS:

$$\mathrm{RSS} = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 = \sum_{i=1}^{n} \left(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots \hat{\beta}_p x_{ip} - y_i\right)^2$$

- (Optional) In matrix form[1], when $\mathbf{X}^\top \mathbf{X}$ is invertible:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \qquad \text{where} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

[1] This can be derived by setting the partial derivatives of RSS to 0, but is beyond the scope of STA 35C

Pop-up quiz: Interpreting coefficients

Scenario: We fit a multiple linear regression model on the `Advertising` dataset:

$$\texttt{Sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{Radio} + \beta_3 \texttt{Newspaper} + \varepsilon.$$

Suppose that we obtain: $\hat{\beta}_1 = 0.04, \quad \hat{\beta}_2 = 0.18, \quad \hat{\beta}_3 = -0.02.$

Question: Which statement *best* describes the meaning of $\hat{\beta}_1 = 0.04$ in this model?

Multiple-choice answers:

A) TV advertising alone explains 4% of the variation in `Sales`

B) A 1-unit increase in `TV` causes `Sales` to increase by exactly 0.04

C) Holding `Radio` and `Newspaper` fixed, a 1-unit increase in `TV` is associated with an average increase of 0.04 units in predicted `Sales`

D) The effect of `TV` is 0.04 only when `Radio` and `Newspaper` are both 0

Answer: C.

In multiple regression, a coefficient measures the average change in the predicted response for a one-unit increase in that predictor, holding the others fixed

Some key questions with multiple predictors

When we perform multiple linear regression, we often want to answer questions like:

- Are predictors $X_1, \ldots, X_p$ related to Y (i.e., do they help predict Y)?
- Which subset of $X_1, \ldots, X_p$ is most important?
- How well does the model fit the data?
- Given new predictor values, what response value should we predict and how accurate is that prediction?

Let's address these questions one by one

Hypothesis testing for relationship between Y and each X_j

Recall from simple linear regression:

$$H_0 : \beta_1 = 0 \qquad \text{vs.} \qquad H_1 : \beta_1 \neq 0, \qquad t = \frac{\hat{\beta}_1}{\widehat{\mathrm{SE}}(\hat{\beta}_1)}.$$

- **Idea:** If H_0 is true, t follows the t-distribution and will likely be close to 0
- Thus, a very large $|t|$ value is an evidence against $H_0 \rightarrow$ reject H_0

In multiple linear regression, we do the same, but for each predictor:

$$H_0 : \beta_j = 0 \qquad \text{vs.} \qquad H_1 : \beta_j \neq 0.$$

- This asks whether X_j is associated with Y *after accounting for the other predictors*
- The answer can differ from simple regression if the predictors are correlated
- Also, we may instead ask whether *any* predictor is useful at all

Advertising example: Simple vs. multiple regressions

Q: Is newspaper useful in predicting sales?

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

	Coefficient	Std. error	t-statistic	p-value
Intercept	9.312	0.563	16.54	< 0.0001
radio	0.203	0.020	9.92	< 0.0001

	Coefficient	Std. error	t-statistic	p-value
Intercept	12.351	0.621	19.88	< 0.0001
newspaper	0.055	0.017	3.30	0.00115

Figure: Separate simple regressions suggest TV, radio, *and* newspaper are all significant [JWHT21, Tables 3.1 & 3.3].

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Figure: Multiple regression suggests newspaper is *not* significant [JWHT21, Table 3.4].

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Figure: Correlation matrix for TV, radio, newspaper, and sales [JWHT21, Table 3.5].

Takeaway: In multiple regression, β_j measures the association between X_j and Y *holding the other predictors fixed*; correlated predictors can therefore change significance

`Advertising` example: Testing single vs. any predictor

Q: Is *"any"* of `TV`, `radio`, `newspaper` useful in predicting `sales`?

We now test a different, *joint* hypothesis:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \beta_j \neq 0$$

This can be tested using the *F-statistic*:

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} \qquad \begin{cases} \text{Reject } H_0 & \text{if F is large,} \\ \text{Cannot reject } H_0 & \text{if F is "typical".} \end{cases}$$

Rationale: If H_0 is true, F will likely be close to 1; a very large F is evidence against H_0

- If all predictors are useless, adding them should not reduce RSS much, so F should be near 1
- A large F means the predictors reduce RSS much more than we expect from noise alone
- Under H_0, F follows an F-distribution with $(p,\ n-p-1)$ degrees of freedom

We will discuss how to select "important" predictors later in this course

Evaluating the model fit

The quality of a multiple linear regression fit can be measured by the RSE or the R^2

- **Residual standard error (RSE)**: *"typical size of deviation, in the same units as Y"*

$$\mathrm{RSE} = \sqrt{\frac{\mathrm{RSS}}{n-p-1}} \quad \text{where} \quad \mathrm{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- **The $\mathbf{R^2}$:** "the proportion of variance in Y explained by X"

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \quad \text{where} \quad \mathrm{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

  - R^2 **never decreases** when more predictors are added
  - **"Adjusted"** R^2 compensates for adding predictors:

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)}$$

    - It increases only if the added predictors improve fit to justify the extra complexity

Pop-up quiz: R^2 vs. adjusted R^2

Scenario: Suppose a model with one predictor has

$$R^2 = 0.80, \quad R^2_{\text{adj}} = 0.79.$$

After adding a second predictor X_2 (suspected to be mostly noise), we get:

$$R^2 = 0.82, \quad R^2_{\text{adj}} = 0.78.$$

Question: Why did R^2 go up while R^2_{adj} went down?

A) There must be a calculation error; if R^2 increases, R^2_{adj} must also increase.
B) X_2 reduced RSS a little, so R^2 increased, but the improvement was too small to justify the extra parameter, so adjusted R^2 decreased, hence, R^2_{adj} drops
C) Adjusted R^2 always decreases whenever we add a predictor, no matter how useful it is
D) R^2 does not measure model fit at all, whereas R^2_{adj} is the only valid measure of fit

Answer: B.

R^2 never decreases when predictors are added, whereas adjusted R^2 penalizes extra complexity and can go down if the new predictor adds little real signal

Confidence intervals and prediction intervals

With $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$, we predict $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$

How certain are we about this prediction?

- $\hat{y} = \hat{f}_{\hat{\beta}}(x)$ only *estimates* $f_\beta(x) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$.
- $y = f(x) + \varepsilon$ has an error term, so additional variability.

Confidence interval for the mean response $f(x)$:

- Reflects uncertainty in prediction due to estimating coefficients
- Across repeated samples, 95% of CIs would contain the true *mean response* at x

Prediction interval for a new response $Y_{\textbf{new}} \mid X = x$:

- Accounts for uncertainty in both $\hat{y} = \hat{f}(x)$ *and* the random noise ε
- Across repeated samples, 95% of PIs would contain a new *observed response* $y = f(x) + \varepsilon$
- The PI is always wider than the CI as it includes both estimation uncertainty & random noise

Exact formulas are beyond our scope, but in **R**:

```
predict(model, newdata = x0, interval = "confidence", level = 0.95)
```

What if there is a non-linear relationship?

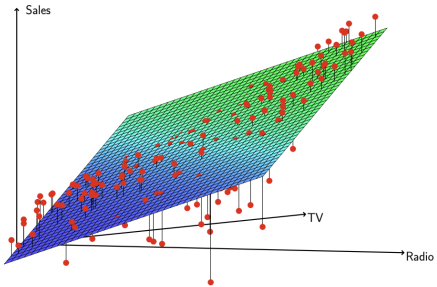

Figure: Pronounced synergy between `TV` and `Radio`; positive residuals cluster along the 45-degree line [JWHT21, Figure 3.5].

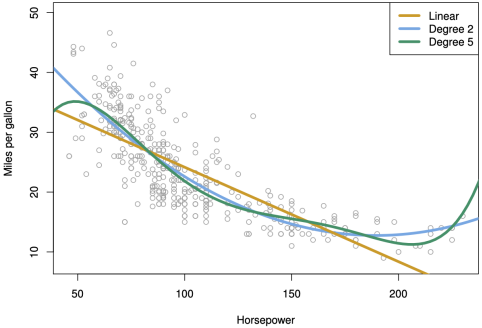

Figure: A non-linear relationship between `mpg` and `horsepower` is noticeable [JWHT21, Figure 3.8].

→ We can enrich the model by adding **interaction** terms (`TV` × `Radio`) or **non-linear** terms ($\texttt{horsepower}^2$) to capture these effects

Polynomial regression: Higher-order and interaction terms

Polynomial regression extends the linear model by including powers of predictors[2]:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon$$

- Treated as multiple linear regression on transformed predictors $(X, X^2, \ldots, X^d)$
- Although non-linear in X, the model is still linear in the coefficients β_j

Example: Interaction effect (synergy between `TV` and `Radio`)

$$\begin{aligned}\texttt{Sales} &= \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{Radio} + \beta_3 \underbrace{\texttt{TV} \times \texttt{Radio}}_{\text{interaction term}} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \texttt{Radio})\, \texttt{TV} + \beta_2 \texttt{Radio} + \varepsilon\end{aligned}$$

Example: Quadratic model

$$\texttt{mpg} = \beta_0 + \beta_1 \texttt{horsepower} + \beta_2 \texttt{horsepower}^2 + \varepsilon$$

[2] More generally, $Y = \sum_{\boldsymbol{\alpha}:|\boldsymbol{\alpha}|\leq d} \beta_{\boldsymbol{\alpha}} \mathbf{X}^{\boldsymbol{\alpha}} + \varepsilon$ where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)$ and $\mathbf{X}^{\boldsymbol{\alpha}} = X_1^{\alpha_1} X_2^{\alpha_2} \cdots X_p^{\alpha_p}$

Model flexibility and the risk of overfitting

A more flexible model can fit the training data more closely

- Adding predictors or higher-order terms can only decrease the training RSS
- Training RSS = RSS measured on the training data

But a better fit on the training data does not necessarily mean a better model

- Our actual goal is good performance on new, unseen data
- A model that fits the training set extremely well may perform worse on a test set

Overfitting: If a model is too flexible, it can fit noise as well as signal

- Typical symptom: very small training error, but noticeably larger test error
- We will briefly explore this phenomenon in Homework 2, Problem 4
- We will return to this issue repeatedly throughout the course

Wrap-up

Multiple linear regression extends simple linear regression to several predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

- Each coefficient β_j describes the average change in the mean response for a one-unit increase in X_j, holding the other predictors fixed
- Individual t-tests and the overall F-test answer different questions: Association to one predictor vs. any predictor
- Model fit is summarized by RSE, R^2, and adjusted R^2; adjusted R^2 helps compare models with different numbers of predictors
- Confidence intervals target the mean response, prediction intervals target a new observation
- Interaction and polynomial terms make the model more flexible, but extra flexibility increases the risk of overfitting

Next lecture: Dummy variables, pitfalls in linear regression, etc.

References

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.
Springer, New York, NY, 2nd edition, 2021.